



Preliminary engine design and inlet optimization of the MULDICON concept

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ABSTRACT

In this study, aspects of the propulsion design for a highly swept wing configuration with military application are presented. These include the design of the thermodynamic cycle and the automated optimization of the inlet geometry. First, the thermodynamic design process of a conventional turbofan engine is described, which is intended to meet the requirements of a generic UCAV configuration. In addition to the thrust requirements derived from the mission profile, other constraints such as maximum fuel consumption, available installation space and aerodynamic and structural limits dimension the model in preliminary design. The resulting engine data are used as boundary conditions for the CFD simulation of the associated engine intake. This CFD calculation is part of an optimization process chain for determining the inlet duct geometry with low total pressure loss and low engine visibility. In order to estimate the effects of the different total pressure losses on the mission fuel consumption, the results obtained using CFD calculations are fed back into the performance simulation. This allows quantifying the influence of the inlet geometry on the global engine parameter. Based on the investigations and obtained results, the final configurations for the engine and intake are selected.

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1. Introduction

Detectability is an important design aspect of military aircraft operating in enemy airspace. The use of various methods and technologies in the preliminary design process can reduce the radar and infrared signature and thus, lower the detectability. Sharp-edged contours and radar absorbing materials are for instance important methods to reduce the radar signature of the aircraft outer shape. For this reason, fuselages with low observable (LO) characteristics have a special angular shape. These methods are insufficient for the engine, which is a major signature source besides the aircraft fuselage. Therefore engine integration in combination with curved inlet and outlet ducts is a very effective method to considerably reduce the infrared, radar and acoustic signature emitted by the engine [1]. However, such integration is associated with special requirements that affect the preliminary design of the propulsion system.

In this study, selected aspects of these requirements are presented using the example of engine predesign for a generic UCAV configuration with LO characteristics. The configuration, called MULDICON, is a highly swept flying wing aircraft. It was designed

and studied in a multidisciplinary design process by the NATO research panel AVT-251 [2,3].

The aspects presented here relate to the thermodynamic preliminary design of a conventional turbofan engine taking into account the engine integration, a multi-objective optimization of the intake geometry as well as the interaction between these components.

2. Engine design

The choice of a turbofan engine as propulsion concept of the UCAV configuration is based on the design specifications of the aircraft configuration and the demand for LO characteristics. The dimensioning of the turbofan engine is part of the thermodynamic engine modeling. The aim of this is to determine the component parameters for a desired operating behavior of the engine that can meet the design requirements of the aircraft.

2.1. Engine requirements

The advantage of UCAVs, not having a pilot on board, qualifies them especially for military missions in enemy airspace, such as “suppression of enemy air defenses” or “intelligence, surveillance and reconnaissance”. The associated requirements for the aircraft configurations are:

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Nomenclature

Abbreviation

AIP	Aerodynamic interface plane
BPR	Bypass ratio
DLR	German Aerospace Center
FAR	Fuel-to-air ratio
GTLab	Gas turbine laboratory
HPT	High pressure turbine
HPC	High pressure compressor
ING	Operating point Ingress
LO	Low observability
LPT	Low pressure turbine
MM	Meta-model
MFC	Mission fuel consumption
MULDICON	Multidisciplinary configuration
OPR	Overall pressure ratio
SAS	Secondary air system
TO-EOF	Take-Off end of field
TSFC	Thrust specific fuel consumption
UCAV	Unmanned combat aerial vehicle
UDP	User defined process chain

Symbols

DC60	Distortion coefficient
c	Flow velocity
E	Glide ratio
g	Gravity
$\dot{m}$	Mass flow
Ma	Mach number
p	Total pressure
p_s	Static pressure
s	Mission segment distance
T	Total temperature
T_s	Static temperature
V	Visibility value
y^+	Dimensionless wall distance
ν	Hub-to-tip ratio
Π	Total pressure ratio
ρ	Density
η_{is}	Isentropic efficiency
$\bar{\Psi}$	Average loading factor
ω	Total pressure loss coefficient

- LO properties, in order not to be detected by radar or infrared surveillance
- low-level flight at high speed, to fly underneath enemy radar
- large operating range.

These in turn result in the following requirements for the propulsion system:

- low exhaust temperatures and speeds, to keep the infrared signature low [4]
- small fan diameter, to keep the aircraft flat and the radar cross section of the engine low
- high power density, to enable the aircraft for fast take-off and for flying with high subsonic Mach numbers at low flight level
- low specific fuel consumption (TSFC), to increase mission range.

Together, these requirements result in important goals for engine design. Some of them are contradictory and lead to conflicting objectives in the engine cycle design. Thus, the task of the engine design is to find the best possible compromise between these objectives. The more precise specification of these requirements is made by the selection and definition of the design mission, presented in Fig. 1.

Five operating points are derived from the mission. These operating points have different requirements with regard to engine design (Table 1).

2.2. Performance modeling

2.2.1. Performance synthesis

Performance synthesis is an important element in the preliminary design of engines. In this case, the operating behavior of the selected engine concept is determined with the help of computer-based calculation methods. The main components of the engine are replaced by virtual modules and their operating behavior is represented by analytical formulas and maps. The program used here is DLR's own simulation tool for performance synthesis GTLab Performance [6]. This tool can be used for the thermodynamic simulation of different gas turbine configurations. Fig. 2 schematically shows

the module composition of the turbofan engine in GTLab performance.

2.2.2. Engine cycle and component technology level

The underlying thermodynamic cycle and the technology level of the engine components have a strong impact on engine performance. By improving component efficiencies or reducing pressure losses, the performance of the entire engine is improved. The same applies to higher-level cycle parameters. For example, an increase in OPR or turbine inlet temperature T_4 leads to higher efficiency of the core engine. The technology level according to today's state of the art is determined with the help of literature sources [7].

The characteristic of the intake in the engine performance model is narrowed to the total pressure ratio Π_1 . This is described in the public literature as 99% for often used Pitot intakes at subsonic flight speeds [8]. However, the integrated engine for the UCAV has an inlet duct which will have a lower value due to its S-shape and length. Since this inlet is not designed at the time of the performance calculation, 96% is set as initial value. This value will be updated later in the study by a total pressure loss determined by CFD calculations. A detailed list of all selected component efficiencies and cycle process parameters are described more in detail in [9].

2.3. Engine performance optimization

2.3.1. Performance optimization task

Due to the large number of requirements in engine design, conflicting objectives may arise if two or more requirements are contrary. At this point in the design it is necessary to decide which compromise is best suited. In the predesign of the integrated engine, there is such a conflict in the dimensioning of the fan diameter between the objectives of low visibility and high efficiency. Besides the maximum diameter limit, caused by the available installation space, smaller diameters offer the advantage of greater flexibility in the geometry design of inlet duct and nozzle. This is particularly important as this flexibility can improve the LO characteristics and thus the survivability of the aircraft. However, a reduction of the fan diameter has a negative effect on the TSFC and thus on the mission range [9]. With the help of an automated multi-objective optimization, the most auspicious engine

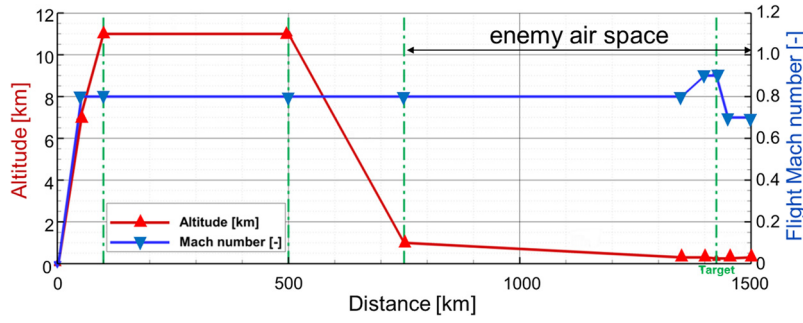


Fig. 1. Generic UCAV design mission [5]. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 1

Operating points of the design mission.

Operating point	Ma_0 [-]	Altitude [m]	$F_N/F_{N,TO}$ [-]	Critical design parameter
TO-EOF	0.2	0	1	therm. stress turbine
Max. Climb	0.8	11000	0.25	max. $\dot{m}_{red}$
Cruise	0.8	11000	0.12	low TSFC
Ingress	0.8	300	0.40	low observability
Dash	0.9	300	0.60	max. thrust

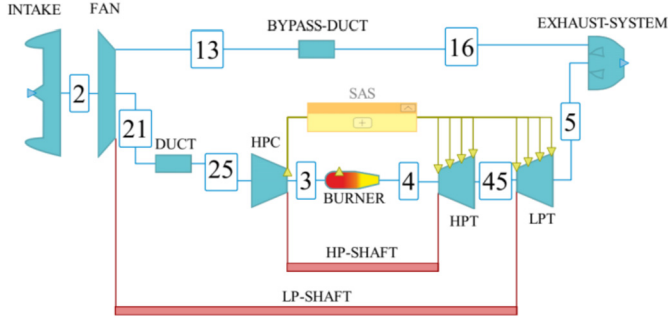


Fig. 2. Performance model of a 2 shaft mixed turbofan engine.

parameters are searched using the following objectives: small fan diameter D_{Fan} and low mission fuel consumption MFC. The engine design variables of the optimization are Π_{HPC} , $\Pi_{Fan,Core}$, BPR and FAR_{Burner} for the design point TO-EOF.

In order to rapidly estimate the fuel consumption m_f of each mission segment i , the converted Breguet range formula is used. The sum of all segment fuel consumptions give the MFC:

$$MFC = \sum_{i=1}^i m_{f,i} = \sum_{i=1}^i m_i \cdot \left(1 - \exp\left(\frac{s \cdot TSFC \cdot g}{E \cdot v}\right) \right). \quad (1)$$

The second objective, fan diameter, is a function of the following parameter at the fan inlet section:

$$D_{Fan} = f(Ma_2, T_{s,2}, p_{s,2}, v_2, \dot{m}_2). \quad (2)$$

$T_{s,2}$ and $p_{s,2}$ are defined by the operating point condition altitude and the flight Mach number. The Mach number at the entrance of the fan Ma_2 and the hub-to-tip ratio v_2 are design specifications which remain constant during the optimization process. Thus, the fan diameter depends only on fan mass flow rate $\dot{m}_2$.

2.3.2. Optimizer AutoOpti

The optimization process is executed with the DLR in-house tool AutoOpti. This is a meta-model assisted framework for single or multi objective optimization, which has been applied for many optimization processes since 2002 [10]. The basic flowchart of the optimizer is shown in Fig. 3. For parallelized optimization there are

Optimization - MM

Optimization - UDP

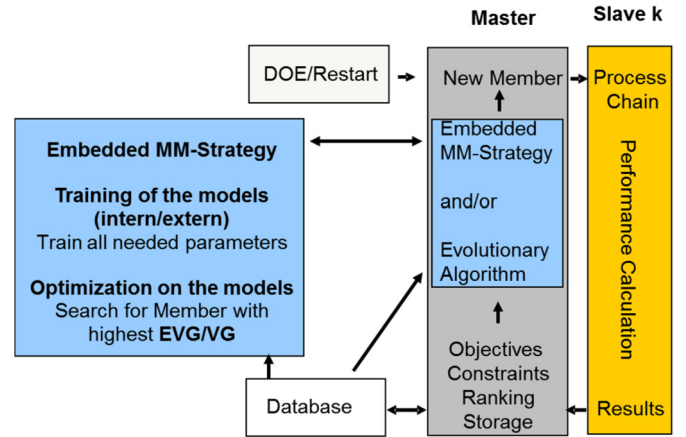


Fig. 3. Flowchart AutoOpti.

one master-process (gray/blue in Fig. 3), handling the optimization algorithms, and several slave processes (yellow in Fig. 3), which are responsible for running the specific user defined process chain (notation: UDP). To calculate the objectives and constraints of a given set of free variables of the optimization process (i.e. called a member), the master process hands over a member to a slave process and receives the simulation results from calculations of the UDP. The calculated objectives and constraints are ranked and stored in the database and afterwards the Pareto rank of all stored members will be updated. In the next step, the master process creates a new more or less auspicious member via evolutionary algorithm or meta-models (notation: MM). In the presented investigation first a design of experiments (DOE) is performed. Afterwards most of the members were created using the embedded MM-strategy by using ordinary kriging models.

2.3.3. Performance optimization results

The goal of the optimization is to find a spectrum of the best possible compromises between the two objectives, which allows the designer to select a highly suitable engine design from the set of best solutions. The optimization is carried out for two different low pressure turbine (LPT) configurations, one-staged and two-staged. These differ only in the restricted values of the load parameter $\bar{\Psi}_{Turbine}$

$$\bar{\Psi}_{LPT} = \frac{2 \cdot \Delta h_t}{\sum u_{mid}^2} \quad (3)$$

and total pressure ratio Π_{LPT} . Both parameters are set to 4.5 for a single-staged or to 9.0 for a two-staged configuration. The best results of the two optimizations are plotted in Fig. 4 over the two objectives. As additional design information the members are dyed

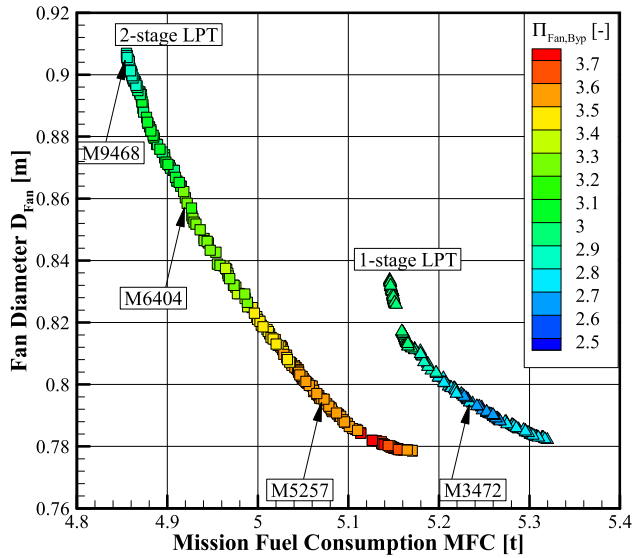


Fig. 4. Best members of the engine performance optimization.

according to the fan bypass pressure ratio at TO-EOF. With decreasing fan diameter the fan pressure ratio of the engine models tends to increase. This can be ascribed to the thrust requirement. To generate the required thrust with lower mass flow, a higher flow velocity at nozzle exit is needed. The higher the fan pressure ratio, the higher the exhaust velocity. For this reason fan diameter and fan pressure ratio have an inverse tendency.

The two point clouds represent the Pareto fronts of the two optimization series. Considering the objectives fan diameter and MFC, each Pareto member represents an optimal compromise. For the selection of the final engine design, further assessment criteria such as the number of fan stages or LO properties are introduced. To determine the optimum engine settings, promising members are selected from the Pareto front and listed in Table 2 according to the chosen assessment criteria. This list is the basis for the decision making of the final engine setting.

The fan or LPT stage numbers are usually important design parameters in preliminary design, as more stages imply higher weight and acquisition costs. For this reason, they are also selected as assessment criteria. However, they play an informative role here, since the design focus of the MULDICON Configuration is on the very low observability properties. Engine parameters that have a large impact on the observability are nozzle exit temperature and nozzle exit speed ($T_{8,ING}$, $v_{8,ING}$) [4]. In contrast to the other assessment criteria, these parameters are listed for the reconnaissance-critical operating point Ingress.

In consultation with the design disciplines of overall-design, infrared signature and integration of intake and nozzle, member M9468 was selected as the final engine configuration, due to its good LO characteristics and the low MFC.

2.4. Engine annulus estimation

After determining the optimal settings of the engine performance model, the components can be geometrically sketched based on these parameters. The engine preliminary design environment GTlab offers the possibility to transfer the results from the performance calculation into the module GTlab-Sketchpad [11] and to perform a geometric preliminary design there. In GTlab-Sketchpad, first turbomachinery annulus, engine dimensions and system weights are determined using fast calculation methods. The input values required for these calculations are design decisions based on literature data [7]:

Table 2

Assessment criteria of the selected member.

Assessment criteria	M9468	M6464	M5257	M3472
Objectives				
MFC _{rel} [%]	+1.19	+2.59	+5.52	+8.99
D_{Fan} [m]	0.908	0.860	0.796	0.796
Weight and costs				
No. Stages Fan	2	2	2	2
No. Stages LPT	2	2	2	1
LO properties				
$T_{8,ING}$ [K]	577	612	692	692
$c_{8,ING}$ [m/s]	439	452	480	480

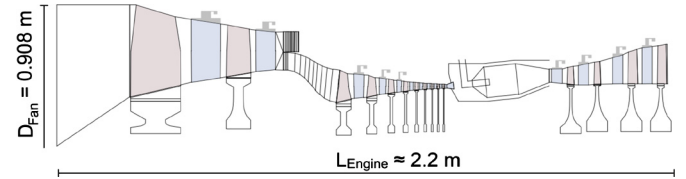


Fig. 5. Annulus estimation of member M9468.

Table 3

Resulting engine design parameters of M9468.

Parameter	Unit	Condition	Value
Thrust	kN	TO	60
OPR	–	TO	30.5
$\dot{m}$	kg/s	TO	114
T_4	K	TO	1740
TSFC	g/kNs	Cruise	23.8
BPR	–	Cruise	1.7
Length	m	–	2.2
No. Stages HPC	–	–	9
No. Stages HPT	–	–	2

- hub-to-tip ratios or “radius rules”,
- axial in- and outflow Mach numbers,
- compressor and fan tip speeds and
- average stage loadings.

Thus, the Sketchpad processes enable a direct evaluation of the influences of the cycle process on the relevant characteristics of the overall system. The result of this design process, shown in Fig. 5, serves a placeholder for the engine and as a boundary condition for the intake and nozzle geometry design.

The determined performance and geometry information of member M9468 serve as boundary conditions in the subsequent inlet geometry design. The aim of this design is to generate an inlet duct with the lowest possible total pressure loss, so that the negative impact on fuel consumption is limited. At the same time, the inlet duct has the task of reducing the visibility of the aerodynamic interface plane (AIP) at fan inlet. This is achieved by a curved shape of the duct, which tends to increase the total pressure losses and the inhomogeneity of the engine inflow. Using CFD calculations, this duct is optimized with regard to the objectives named above. The preparation, execution and evaluation of this engine inlet duct optimization are described in the following chapter. The final engine design data derived from performance optimization and following annulus estimation are listed in Table 3.

3. Inlet duct optimization preparation

3.1. Flow solver and boundary conditions

The CFD flow simulations are performed in the flow solver TRACE 9.0, which is developed by the DLR Institute of Propul-

sion Technology in cooperation with MTU Aero Engines [12]. With TRACE the Reynolds averaged Navier-Stokes equations are discretized using the finite volume method to calculate stationary and transient states of flows in axial and radial turbomachinery components. In the presented investigations the two equation $k-\omega$ model by Wilcox is used for closing unknown turbulent terms [13].

Within the optimization process the calculations are performed with fully turbulent flow under stationary conditions. Furthermore a homogeneous and parallel inflow state is assumed, as well as the influence of the spinner at the compressor front face is neglected. Additionally, the fuselage boundary layer thickness is not considered. The wall treatment is done with low Reynolds. Therefore the dimensionless wall distance y^+ is nearly 1. To ensure fully developed boundary conditions, the mesh includes extrusions at the duct entry and exit area.

For running the CFD simulations the total pressure and total temperature as well as turbulence parameters of the $k-\omega$ model are set at the inflow panel using Riemann boundary conditions. The inlet boundary conditions are set by the performance optimization in 2.3.3 at the operating point Cruise. As outflow boundary condition a constant mass flow ratio is required. The off-design CFD simulations are performed with the data for operation point Ingress.

3.2. Objectives and process chain

For the intake optimization process, also executed with AutoOpti, it is necessary to calculate objectives to get a quantitative assessment of each generated inlet duct. One main task of the S-shaped duct is to provide a stable, low-loss and homogeneous flow to the AIP area. The total pressure losses are quantified by usage of the total pressure loss coefficient

$$\omega = \frac{p_1 - p_2}{\rho_1 c_1^2 / 2} \quad (4)$$

by determining the difference between the S-duct entry and exit total pressure, in relation to the dynamic pressure at the entry. Additionally, the total pressure ratio

$$\Pi_1 = \frac{p_2}{p_1} \quad (5)$$

is used to quantify the total pressure losses, comparing the exit total pressure p_2 with the entry total pressure p_1 . As second objective the quantification of the AIP-visibility seen from the S-duct entry is necessary. Therefore the AIP area is discretized in a number of points D_{total} . The number of these points visible from at least one point of the discretized entry surface is called D_{vis} . The ratio between visible and total amount of points is defined as visibility value:

$$V = \frac{D_{\text{vis}}}{D_{\text{total}}} \quad (6)$$

Both, the total pressure loss coefficient ω and the AIP-visibility V have to become minimized within the optimization series. Additionally, the DC60 value provides information about the flow homogeneity at the AIP area by comparing the averaged total pressure p_m with the minimal total pressure $p_{60,\text{min}}$ in a 60°-segment, normalized by the averaged dynamic pressure of the S-duct intake [14].

$$\text{DC60} = \frac{|p_m - p_{60,\text{min}}|}{\rho c_1^2 / 2} \quad (7)$$

The used process chain within a slave process, to vary 27 free geometrical parameters in AutoOpti, is shown in Fig. 6. First, the geometric parametrization of a member is checked with geometrical

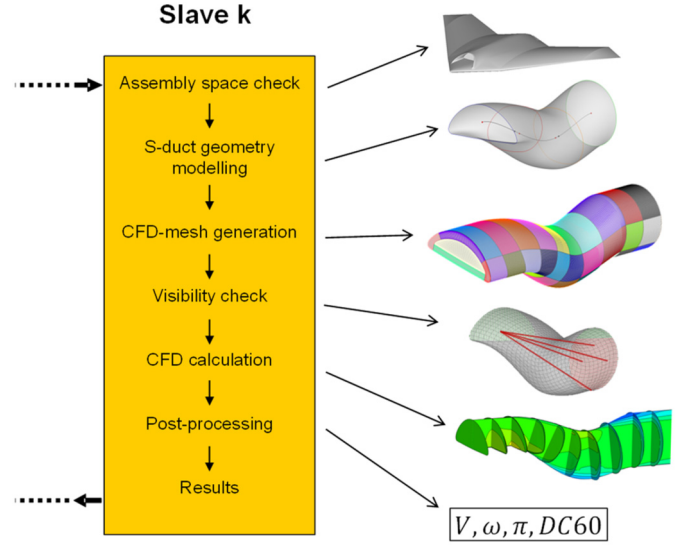


Fig. 6. Used process chain in AutoOpti.

data of the UCAV assembly space. Afterwards the S-duct geometry and CFD mesh creation are performed using the DLR inhouse tool DuctGen [15]. Afterwards the AIP-visibility value V of the AIP area is calculated in the visibility-tool, by checking the number of visible points in the AIP. Detailed information about the used tools are described in [16]. In the last step the CFD flow simulation is carried out followed by post processing.

4. CFD results

This chapter presents the results of the optimization process for the operation point Cruise. Some more detailed investigations in this operating point are shown in [16]. Based on these results, a selection of auspicious configurations is made. These members are compared in AIP-visibility, global flow field and flow homogeneity in the AIP area (Section 4.3.). Additionally, an evaluation of the aerodynamic off-design performance is shown.

4.1. Optimization results

With the convergence of 2525 members the optimization series in the design point Cruise is aborted. Fig. 7 shows the resulting Pareto front with the values of the total pressure loss coefficient and the AIP-visibility plotted against each other. Moreover the DC60 values are shown as color plot.

As can be seen from the Pareto front a reduction in visibility leads in general to an increase in total pressure loss and in DC60, i.e. in flow inhomogeneity at the AIP area. Additionally, the total pressure losses tend to increase drastically below a visibility value of around 30%, as well as the DC60 values are strongly increasing below a visibility value of 25%. However, there are some outliers in DC60, due to the total pressure averaging in the 60° segments.

All members in the Pareto front form a spectrum of compromises between flow losses and the AIP-visibility. For further analysis of the AIP-visibility and flow phenomena five members are selected whose results are shown in Table 4.

4.2. Performance model update

For a better evaluation of the inlet geometry, the calculated total pressure losses are converted into mission fuel consumption using the performance calculation. Thus, the impact of the inlet geometry on the MFC can be quantitatively evaluated. Therefore the total pressure losses are transferred to the performance model

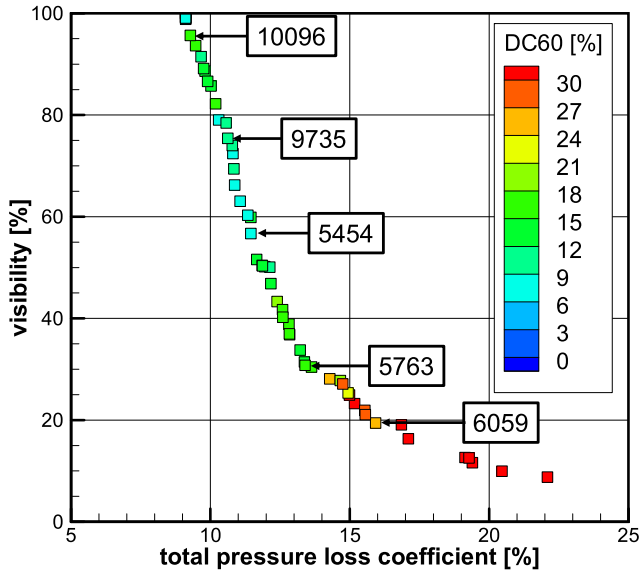


Fig. 7. Resulting Pareto front.

Table 4
Results for operating point Cruise.

Member	10096	9735	5454	5763	6059
V [%]	95.7	75.4	59.7	31.4	19.4
ω [%]	9.3	10.6	11.5	13.4	15.9
Π_1 [%]	98.0	97.7	97.5	97.1	96.5
DC60 [%]	15.4	10.4	7.5	14.1	24.1

Table 5
Results for operating point Ingress and MFC.

Member	10096	9735	5454	5763	6059
V [%]	95.7	75.4	59.7	31.4	19.4
ω [%]	10.0	11.1	11.6	13.1	15.1
Π_1 [%]	98.6	98.5	98.4	98.2	97.9
DC60 [%]	10.9	7.9	6.1	12.6	26.2
ΔMFC [%]	0	+0.188	+0.221	+0.339	+0.684

and used instead of the initial total pressure ratio of $\Pi_1 = 96\%$. How this parameter affects the thermodynamic parameters of the engine model is described hereafter.

Higher pressure losses, represented by lower Π_1 , results in lower OPR (overall pressure ratio), which is the pressure ratio between ambient condition and compressor outlet. An increase in OPR has a positive effect on the thermal engine efficiency and thus, on fuel consumption. A reduction of inlet losses results in a reduction of MFC. The selected inlet geometries in Table 5 show this correlation of Π_1 and MFC.

In performance calculations it is usually assumed that the inlet total pressure ratio is the same for all operating points. To reduce the inaccuracy caused by this assumption, the presented CFD results are used to determine Π_1 for the operating point Ingress. With a distance of 1200 km, Ingress is the mission segment with the largest impact on the MFC (total mission range 3000 km), followed by Cruise with 800 km. The CFD boundary conditions, derived from the performance calculations, are adapted to the Ingress operating condition.

These changed boundary conditions result in a lower total pressure loss of the inlet, in particular due to a lower reduced mass flow. Under the assumption that this has more or less the same effect on each inlet geometry, it is further assumed that the order of the members in the Pareto front is still the same.

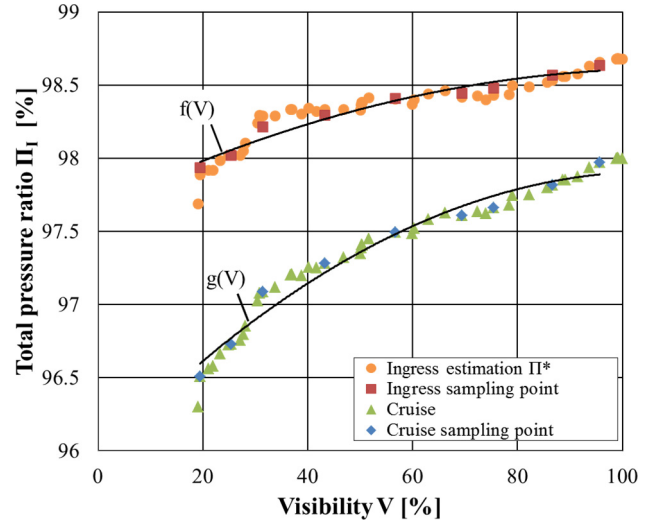


Fig. 8. Total pressure ratio estimation of operating point Ingress.

To verify this assumption, the inlet losses for 20% of the Pareto members are calculated with CFD and Ingress boundary conditions. The Results show a very similar trend as in Cruise. To reduce computational effort, this assumption is used to infer the total pressure loss in Cruise from the total pressure loss in Ingress using a functional relation between Π_1 and visibility. This relationship, formed by regression is shown in Fig. 8 and denoted as $f(V)$. The difference to a polynomial function for Cruise, also determined by regression and denoted as $g(V)$, is used to estimate the Π_1 offset between Cruise and Ingress.

$$\Pi_{1,ING} = \Pi^* = \Pi_{1,CR} + (f(V) - g(V)) \quad (8)$$

The calculation of the MFC is carried out with the help of the total pressure losses in Cruise and Ingress. The results are shown in Fig. 9. These show that the difference between the intakes with high and low visibility is only 80 kg in MFC. Due to this, the total pressure loss over the inlet duct is minor impacting on the selection of the final inlet geometry. Other parameters such as visibility and a homogeneous inflow (DC60) at AIP have a greater influence on the decision. Therefore further investigations of the visibility values and the flow solution are the more important factors to decide for a final geometry.

4.3. AIP-visibility analysis

As can be seen by the S-duct centerline in Fig. 10 (top), the AIP-visibility is reduced by the distances between the duct entry maximum respectively the duct exit maximum and the duct minimum. Thus, decreasing the duct minimum and increasing the duct exit maximum lead to lower values of the AIP-visibility.

The fact that member 5763 has 12% higher visibility than member 6059 is declared by the relative area distribution referred to the duct inlet area (Fig. 10, bottom). By reducing the local area size after a maximum at a relative duct length of 25%, the following minimum reduces the AIP-visibility by locally reducing the area to be viewed through. Hence this effect member 6059 reaches a lower visibility compared to the other investigated members.

4.4. Design point flow field results

In Fig. 11 (top) the Mach number distribution of member 5763 is shown exemplarily for all members, due to slight differences in the global flow phenomena. For all investigated members, a main

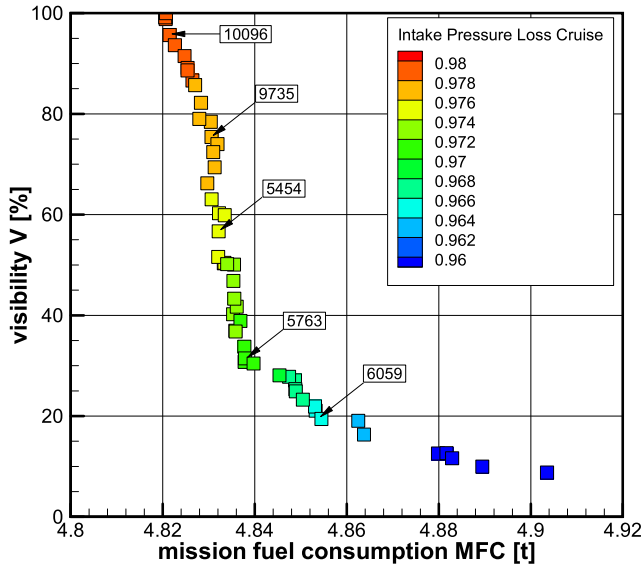


Fig. 9. Comparison of visibility and MFC for all intake geometries of the Pareto front.

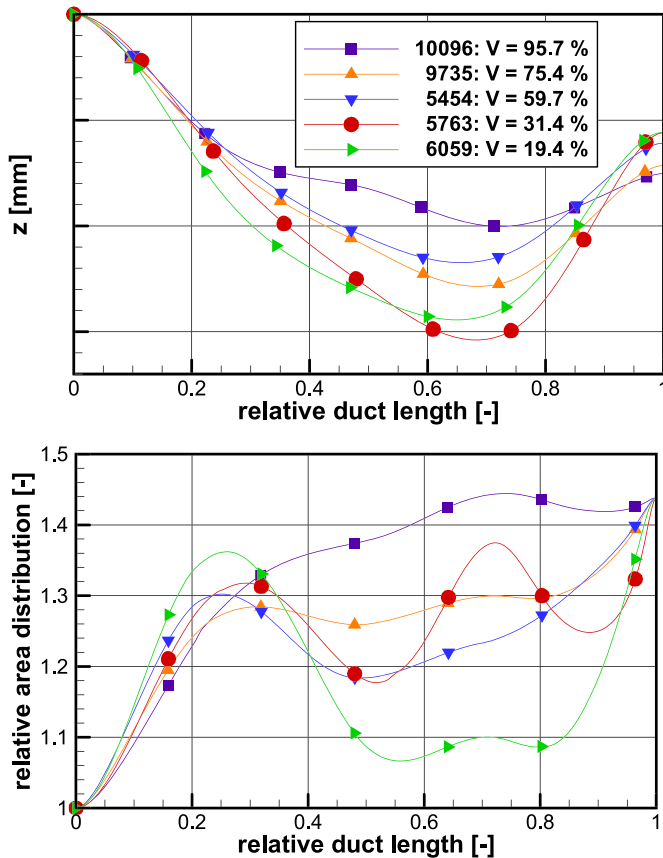


Fig. 10. S-duct centerline (top) and area distribution (bottom).

flow with low losses is observed. Due to the overall area expansion between the duct entry and exit, the inflow Mach number is reduced. Furthermore there are existing local flow decelerations and accelerations, because of the local area expansions and contraction seen in Fig. 10. Additionally, areas of large flow deflections are identified as a result of sharp bending of the centerline. At the S-duct lower side at entry and exit the flow is accelerated. Between these areas the flow gets decelerated at the inner duct lower side. The opposite effects of flow acceleration and deceleration oc-

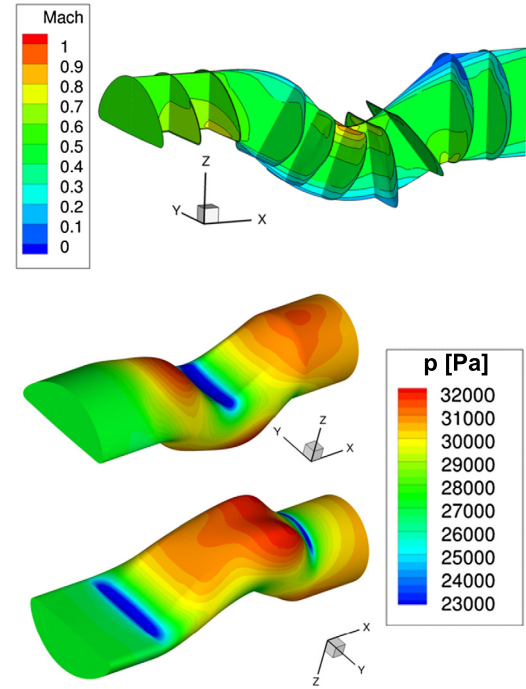


Fig. 11. Mach number distribution (top) and static wall pressure distribution (bottom) of member 5763.

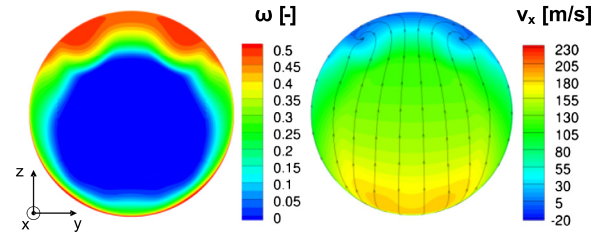


Fig. 12. Local total pressure loss (left) and x-velocity distribution (right) at the AIP area.

cur at the S-duct upper side. Additionally, the near wall pressure gradient increases in flow direction at areas of flow deceleration (Fig. 11, bottom). Thus, there are existing areas with flow separations in the front of the S-duct exit on the upper side. A detailed analysis of the flow field as well as the investigation of flow separation is documented in [16].

In Fig. 12 the local total pressure loss referred to the S-duct entry total pressure visualized to assess the flow homogeneity. In general the effects are comparable between the investigated members from Table 4. Due to the flow deflection on the S-duct upper side in front of the AIP area, the flow velocity on the S-duct upper side is reduced at the AIP upper side. Moreover vortices are induced and thereby flow losses are increased due to friction. In general, the vortex areas increase with a reduction in visibility. But all in all, tolerable flow inhomogeneities can be observed.

However, the risk of occurring flow separations rise with a higher S-duct bending in front of the AIP area, hence a decrease in visibility induces increasing flow losses.

4.5. Off-design results and comparison

To get a more accurate assumption of the S-duct performance with regard to total pressure losses, CFD calculations are performed in the off design operating point Ingress. The results of these calculations are shown in Table 5. In comparison to the design point data in Table 4, the total pressure loss coefficient

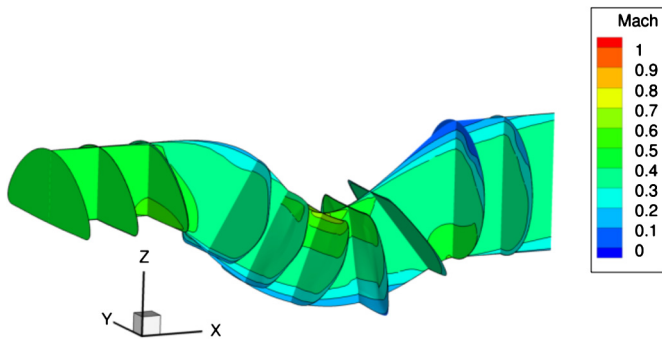


Fig. 13. Mach number distribution of member 5763 in the operating point Ingress.

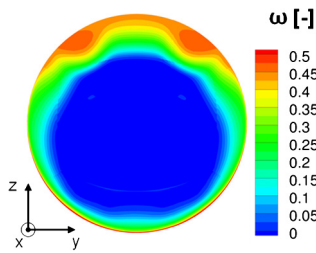


Fig. 14. Local total pressure loss at the AIP area in off design point Ingress.

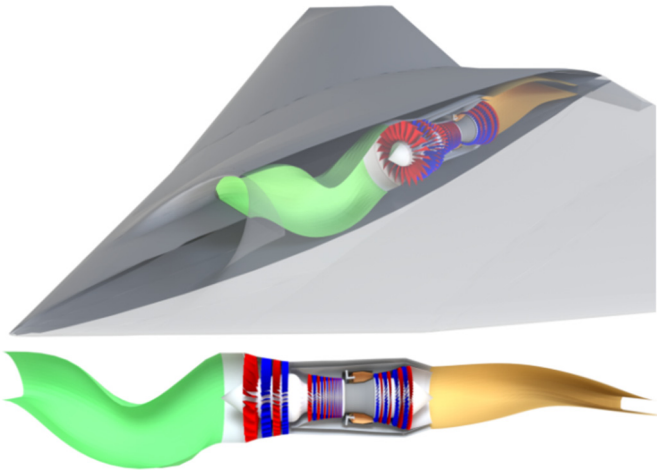


Fig. 15. UCAV diagonally front view (top) with inlet duct and 3D sketch of the estimated engine annulus and nozzle; cross sectional view UCAV power train (bottom).

predicts higher flow losses in off design for high visibility values, but lower flow losses for low visibility values. Comparing the total pressure ratios of the design point with these from the off design point shows, that the overall total pressure losses are decreased. Furthermore the inhomogeneities are reduced for all members except for member 6059.

In Fig. 13 the Mach number distribution of member 5763 in the off design point Ingress is shown. Compared to the flow field in Cruise, the inner flow field Mach number is reduced. Moreover the boundary layer thickness is reduced, due to a higher Reynolds number by operating in Ingress. Thus, the flow deflections at the S-duct bending areas have a smaller impact as in the design point and the total pressure losses are decreased. Furthermore the areas of high total pressure losses in the AIP area are reduced (Fig. 14). Due to this, a more homogeneous inflow to the fan area is provided.

4.6. S-duct integration

Based on the results, the S-duct of member 5763 is selected for engine integration. Since the AIP-visibility is crucially important for the survivability of the MULDICON configuration, some flow losses have to be tolerated. As can be seen from Fig. 15 the S-shaped inlet duct of member 5763 fits into the specified assembly space.

5. Conclusion

In this paper, the design efforts for the propulsion system of a tactical combat drone were presented. These include the modeling of a performance and geometry model of a generic turbofan engine and the geometrical design of the associated S-shaped inlet duct. The requirements and restrictions arising from the design focus of the aircraft were discussed in the first section. In the next step, the thermodynamic performance model with GTlab Performance was created based on these requirements. The requirement for low visibility of the engine AIP was a challenge for the determination of the engine size. A suitable compromise had to be worked out from the conflict of objectives between fuel consumption and fan diameter. Therefore, a multi-objective optimization provided the basis for a multidisciplinary decision. The final engine model, with a fan diameter of $D_{\text{Fan}} = 0.908$ m, was selected for its MFC and low exhaust gas temperature.

Based on this engine model, the next step was to optimize an S-shaped inlet duct. The goal was to reduce the visibility of the AIP while keeping the total pressure loss, resulting from the curved duct, as low as possible. By optimizing the 27 geometric parameters, that define the duct shape, it was possible to show that a significant reduction in the visibility value is reached, considering low total pressure losses. However, a complete concealment of the engine AIP could not be achieved due to the available installation space. A strong increase of the total pressure loss coefficient could already be observed with visibility values of less than 30%. The effect of this total pressure loss on mission fuel consumption was investigated by transferring the CFD results back to the performance model and re-estimating the mission fuel consumption. Since the difference between the lowest and highest consumption is only approx. 80 kg for a mission range of 3000 km, this parameter had a minor influence on the selection of the final inlet duct geometry. Other requirements such as flow homogeneity at the AIP area and reduced visibility were more important in the selection.

Based on the obtained results and further analyses of the flow solutions, the final intake geometry with a visibility value of 31.44% and a total pressure loss coefficient of 13.37% was selected. In combination with the previously designed engine performance model, the mission fuel consumption is estimated to be 4.84 t.

Declaration of competing interest

None declared.

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